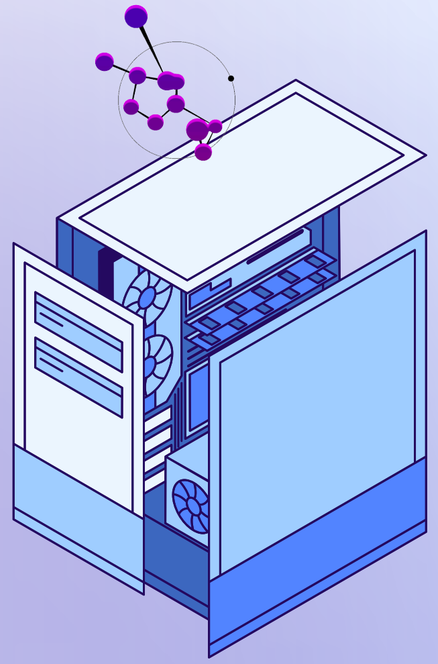
Object-Oriented Library Management System

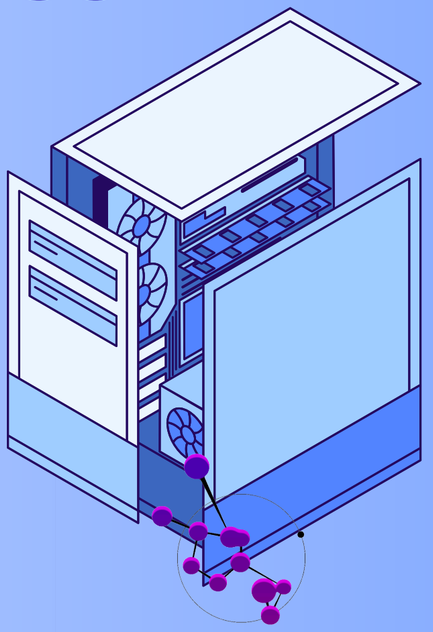
A Complete & Scalable Python Solution with Dynamic GUI

Prepared by: Zahraa Omar Mohamed

Course: Object-Oriented Programming (OOP) - Final Project

Executive Overview & System Purpose

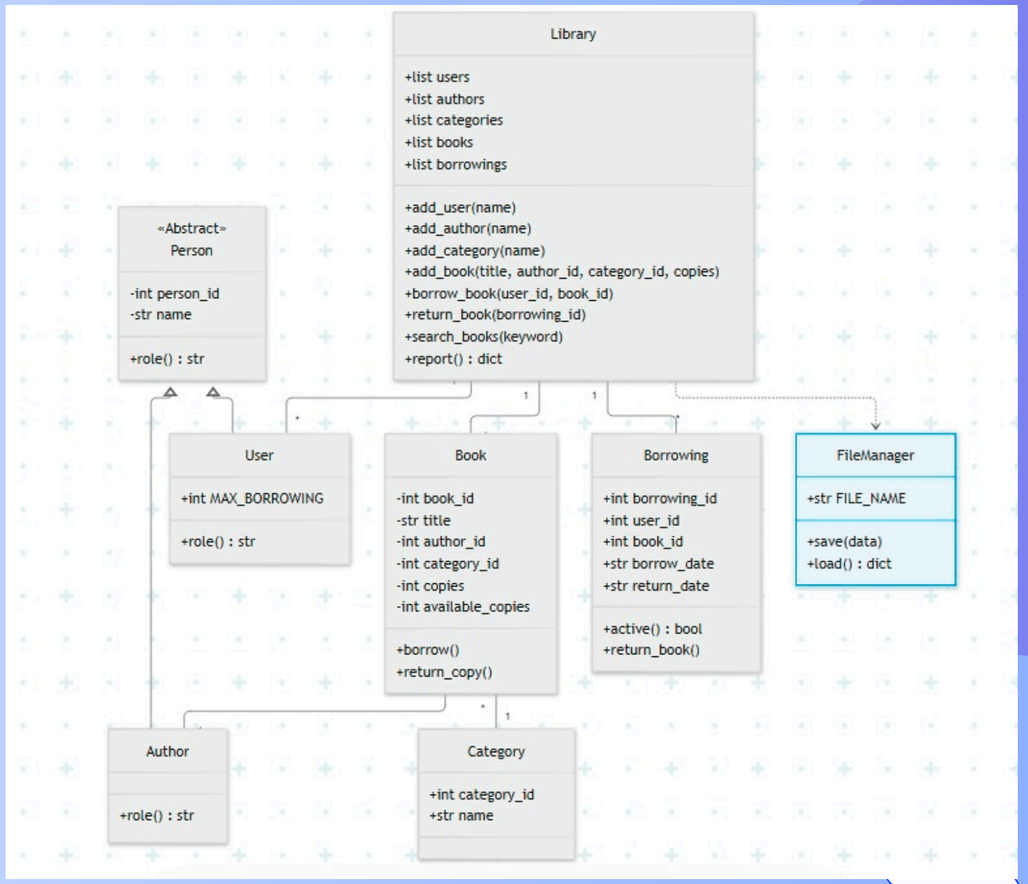




- Designed an end-to-end Library Management System adhering strictly to core Object-Oriented Programming principles.
- Implemented full lifecycle management for Users, Authors, Categories, and Books.
- Enforced real-time borrowing and returning workflows with business logic rules.
- Built with local JSON file persistence and a dynamic Gradio Graphical User Interface

Object Model Architecture

- Abstract Base Class: Person enforces standard attributes and abstract role() contract.
- Inheritance: User and Author extend Person.
- Domain Models: Encapsulated classes for Book, Category, and Borrowing
- Controller: Library acts as the core business logic engine.





Core OOP Pillars

- Abstraction: Person(ABC) defines an abstract template with an abstract role() method.
- Inheritance: User and Author inherit person_id and name from Person.
- Encapsulation: Private attributes (__title, __copies, __available_copies) managed via getters/setters.
- Polymorphism: Overridden role() method returning "User" or "Author" dynamically.

Liceria Tech

AUTOMATED TESTING SUITE (10 SCENARIOS)



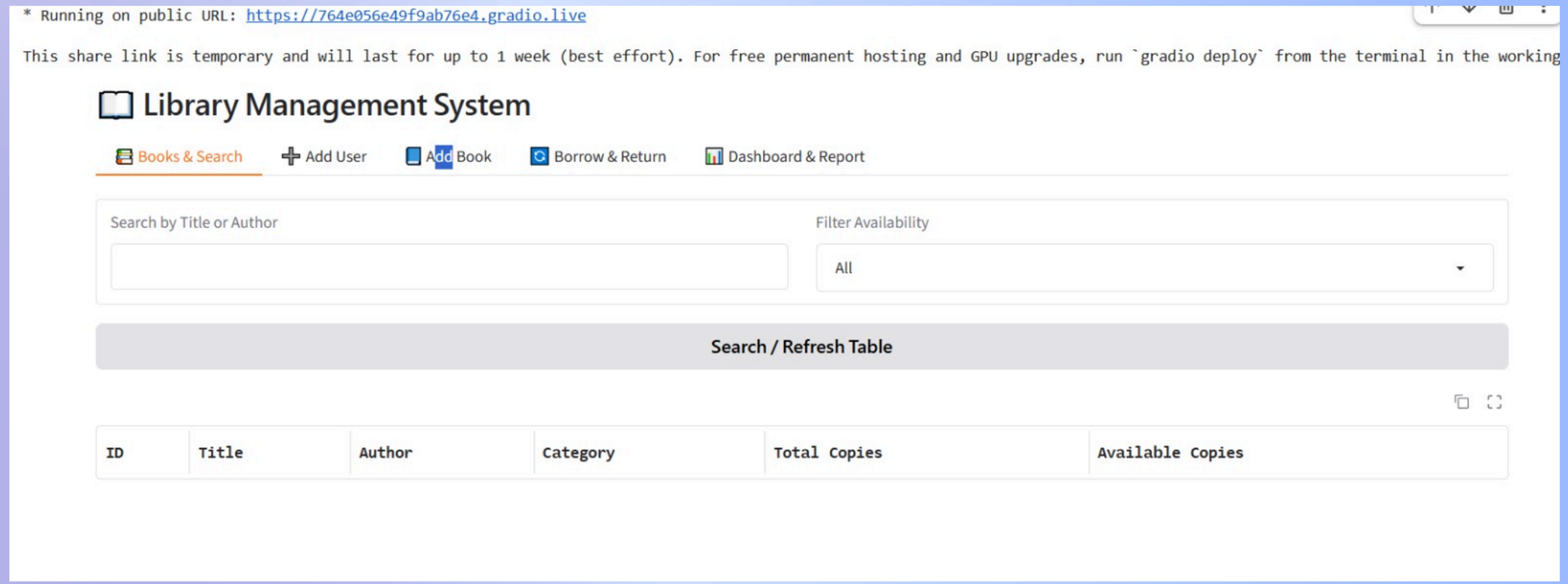
- Successful Workflows (3): Valid book addition, user creation, and borrowing transaction.
- Input Validation (3): Handles empty titles, invalid copies, and non-existent user IDs.
- Business Rules (2): Prevents duplicate book borrowings and enforces max 3-book limit.
- Edge Cases (2): Blocks borrowing when stock is 0 and locks deletion of borrowed books.

```
========== STARTING AUTOMATED TEST SUITE ==========
Test 1 (Add Book): PASS
Test 2 (Add User): PASS
Test 3 (Borrow Book): PASS
Test 4 (Empty Title Validation): PASS
Test 5 (Invalid Copies Validation): PASS
Test 6 (Invalid User Validation): PASS
Test 7 (Borrow Same Book Twice Rule): PASS
Test 8 (Exceed Borrow Limit Rule): PASS
Test 9 (Zero Available Copies Edge Case): PASS
Test 10 (Delete Borrowed Book Edge Case): PASS
========== TEST SUITE COMPLETED ==========
```

NTERACTIVE SEARCH & FILTERING

- Real-time search engine filtering by book title or author name.
- Dynamic dropdown filter to switch between All, Available, or Unavailable books

- Clean interactive interface powered by Gradio framework.



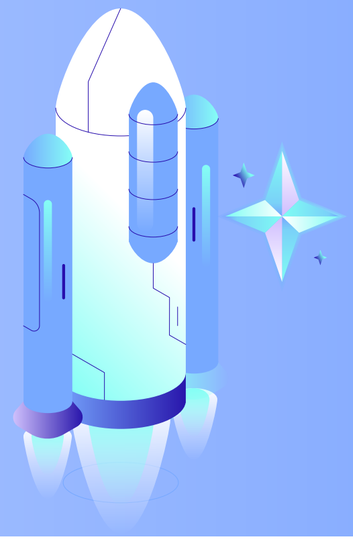
Book Inventory Management

- Structured data grid displaying Book ID, Title, Author, Category, and Stock levels.
- Instant status sync whenever books are borrowed, returned, or newly added.

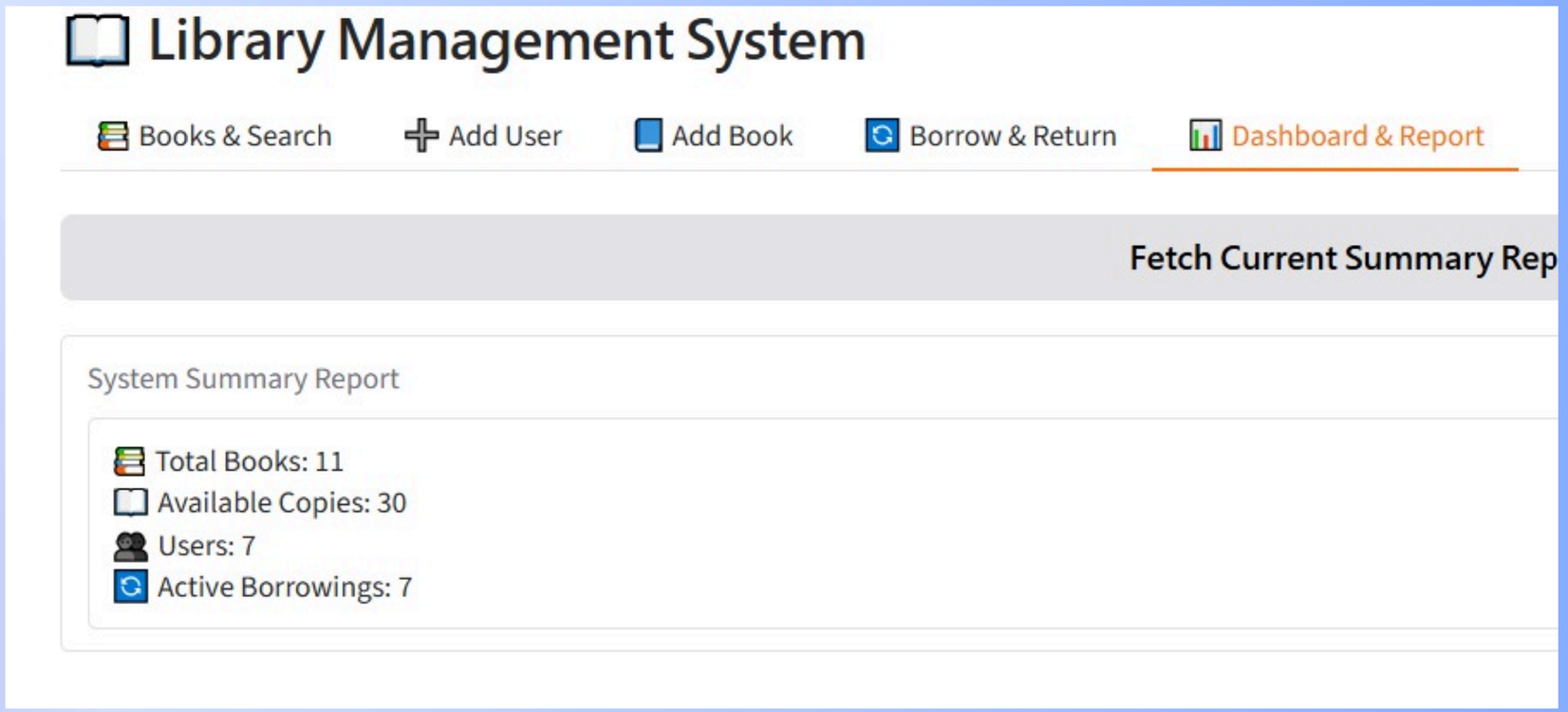
ID	Title	Author	Category	Total Copies	Available Copies
1	Python Basics	Clean Code Author	Computer Science	2	1
2	Book 2	Clean Code Author	Computer Science	2	1
3	Book 3	Clean Code Author	Computer Science	2	1
4	Book 4	Clean Code Author	Computer Science	2	2
5	Rare Book	Clean Code Author	Computer Science	1	0
6	Python Programming	Clean Code Author	Computer Science	1	1
7	Python	Clean Code Author	Computer Science	1	0
8	Python	Clean Code Author	Computer Science	1	0
9	Python 111	Clean Code Author	Computer Science	5	5

Runs · Use via API · Built with Gradio · Settings

Real-time Dashboard & Reporting



- Live Analytics: One-click summary reporting via Library.report().
- Metrics Tracked: Total Books, Available Copies, Registered Users, and Active Borrowings.
- Data Persistence: Automatic saving and reloading via FileManager (library_data.json).



THANK YOU!

PROJECT COMPLETED SUCCESSFULLY